Perhaps this means that the state |*sing*> is somehow incomplete—that there are details of the system that we were sloppy about and didn't measure. After all, earlier we saw a perfectly classical example in which Alice and Bob knew nothing about their coins until they actually looked at them. How is the quantum version different?

In our "classical entanglement" example involving Alice, Bob, and Charlie, it is perfectly clear that there was more to know. Charlie could have sneaked a peek at the coins without changing anything, because classical measurements can be arbitrarily gentle.

Might there be so-called *hidden variables* in the quantum system? The answer is that according to the rules of quantum mechanics, there is nothing to know beyond what is encoded in the state-vector—in the present case, |*sing*>. The state-vector is as complete a description of a system as it is possible to make. So it seems that in quantum mechanics, we can know everything about a composite system—everything there is to know, anyway—and still know nothing about its constituent parts. This is the true weirdness of entanglement, which so disturbed Einstein.

6.9 COMPOSITE OBSERVABLES

Let's imagine a quantum mechanical Alice-Bob-Charlie setup. Charlie's role is to prepare two spins in the entangled state |*sing*>. Then, without looking at the spins (remember, quantum measurements are *not* gentle), he gives one spin to Alice and one to Bob. Although Alice and Bob know exactly what state the combined system is in, they can predict nothing about the outcome of their individual measurements.

But surely knowing the exact state of the composite system must tell them *something*, even if the state is highly entangled. And in fact it does. However, to understand what it tells them, we have to consider a wider family of observables than the ones that Alice and Bob can measure separately, each using *only* his or her own detector. As it turns out, there are observables that can only be measured by using both detectors. The results